Software Engineering

Chapter 6: User Experience (UX) Design & Human-Centered AI

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Chapter 6: Roadmap & Core Curriculum

Part 1: Foundations & Usability Heuristics

- **6.1 Foundations of User Experience (UX):** What is UX, Norman Doors, the 3 bad design archetypes, Taiwan NHI App case study, UI vs. UX distinction, 5-stage Design Thinking.
- **6.2 Nielsen's 10 Usability Heuristics:** System status visibility, real-world match, user control & freedom, consistency, error prevention, recognition over recall, efficiency accelerators, minimalist aesthetics, error recovery, documentation.

Part 2: AI-Powered UX & Interaction Design

- **6.3 AI for UX (AI4UX):** AI as a design amplifier, the RTCF prompt engineering framework (Role-Task-Constraint-Format), automated heuristic audits, multi-agent prototyping.
- **6.4 UX for AI (UX4AI):** Designing for non-deterministic AI, managing latency & thinking traces, human agency & steerability, hallucination guardrails, explainable AI (XAI).
- **6.5 Evaluation & Synthesis:** Heuristic severity ratings (0 to 4), heuristic evaluation vs. user testing, chapter recap quiz, and engineering takeaways.

6.1 Foundations of User Experience (UX)

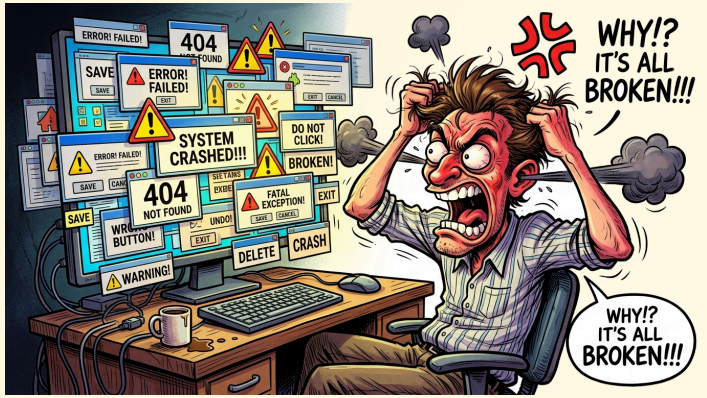
> *"Good design is actually a lot harder to notice than poor design, in part because good designs fit our needs so well that the design is invisible."*
> *— Don Norman, The Design of Everyday Things*

The Classic Anti-Pattern: Norman's Door

- **What is a "Norman Door"?**
  - A door whose design gives false visual clues about how to operate it (e.g., a door with a pull handle that must be pushed, or a flat plate that must be pulled).
- **Affordance vs. Signifier:**
  - **Affordance:** The physical properties and potential actions of an object (a flat surface affords pushing).
  - **Signifier:** Visual cues that signal *where* and *how* the action should happen (e.g., a "PUSH" sign).
- **The Design Axiom:**
  - When an object needs an instruction sign just to open it, its design is fundamentally broken!

The Frustration of Bad UI/UX

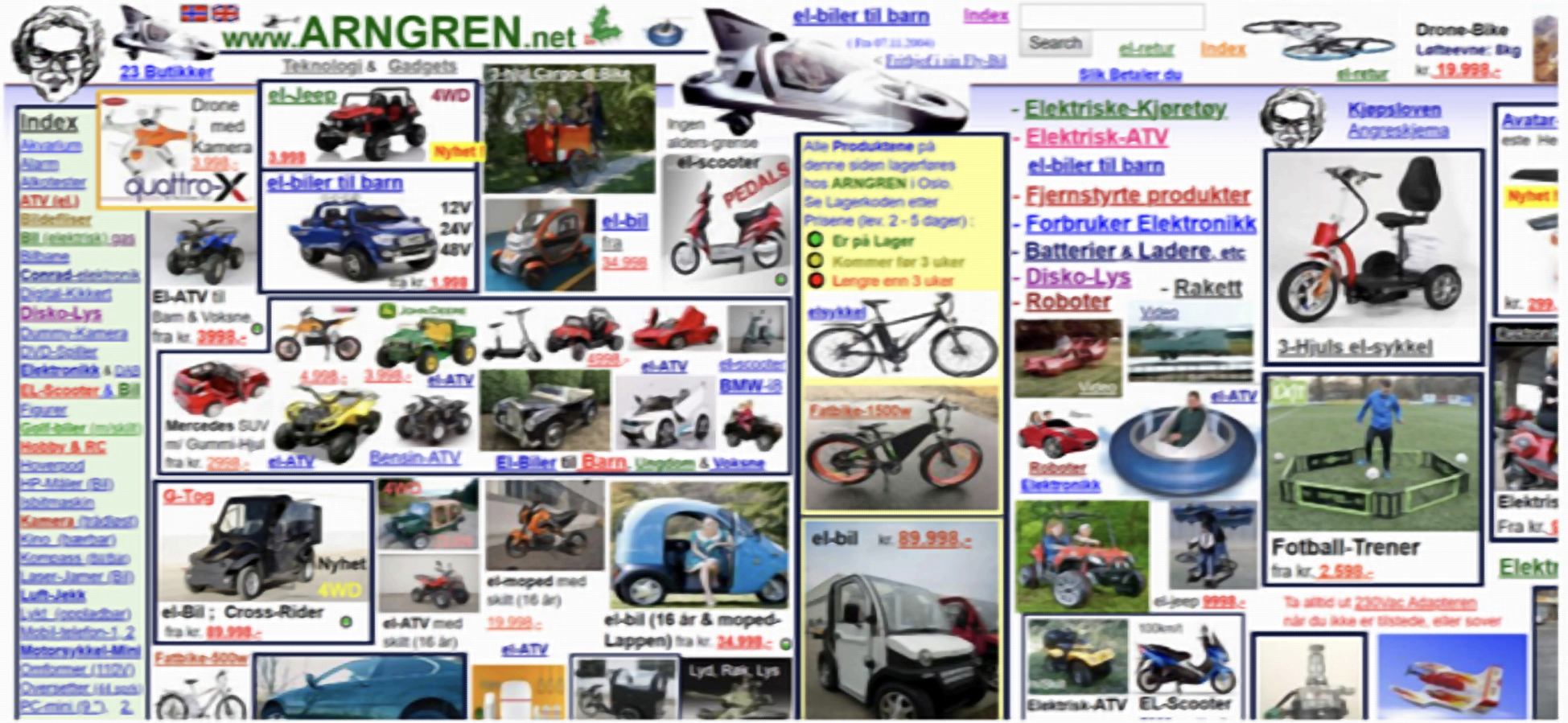
- **The Emotional Toll of Poor Software:**
  - Cognitive overload, confusion, anxiety, and ultimate user abandonment.
  - Users do not blame software architects; they blame themselves or leave 1-star reviews.
- **Why Bad UX Destroys Business Value:**
  - **Cart Abandonment:** 70% of e-commerce carts are dropped due to convoluted checkout flows.
  - **Support Costs:** Thousands of expensive customer tickets originate from confusing navigation.
  - **Employee Burnout:** Enterprise software with terrible UX causes daily operational errors and lost productivity.



3 Classic Bad Design Archetypes

- **1. The "Night Market Chaos" (Arngren-Style Overload):**
  - Stuffs hundreds of products, flashing banners, and conflicting links into a single screen with zero visual hierarchy.
  - The human brain suffers acute cognitive sensory overload.
- **2. The "Paint is Free" (Color & Contrast Nightmare):**
  - Uses neon greens, blazing yellows, and magenta fonts on cyan backgrounds.
  - Violates WCAG color contrast standards; illegible for users with visual impairments.
- **3. The "Where Do I Start?" (Missing Visual Hierarchy):**
  - Clutters dozens of identical, unprioritized buttons across the screen.
  - Fails to establish a primary Call-to-Action (CTA); the user has no idea which button to click first.

Real-World Anti-Pattern: Night Market Sensory Overload



https://www.mockplus.com/blog/post/bad-web-design

Case Study: Taiwan Mask App — The Operational Challenge

- **Context & Operational Urgency:**
  - During the COVID-19 pandemic, Taiwan's National Health Insurance App (健保快易通) handled millions of mask reservations daily.
- **Critical Usability Flaws of the Original Interface:**
  - **Buried Priority:** The critical "Mask Pre-Order" function was hidden inside a cluttered, 20-icon unorganized grid.
  - **Cognitive Search Fatigue:** Elderly citizens and anxious users were forced to hunt across multiple rows of identical icons.
  - **No Visual Hierarchy:** Trivial informational links had the exact same visual weight as life-critical pandemic services.
  - **System Bottlenecks:** Confused users repeatedly tapped wrong options, spiking customer support queues and server latency.

Case Study: NHI Mask App — Before & After Redesign

> *"Interface layout should dynamically reflect urgent user goals, not internal department divisions."*

Before: Bureaucratic Grid Clutter

After: Goal-Directed Hero Flow

Case Study Analysis: Transforming Emergency UX

- **Human-Centered Redesign Strategy:**
  - **Hero Priority Card:** Promoted mask pre-ordering to a prominent top banner occupying the prime mobile thumb zone.
  - **Task Chunking:** Grouped secondary medical services into clear functional cards with clean spacing and high-contrast labels.
  - **Immediate Clarity:** Reduced user decision time from minutes of hunting to a single direct tap.
- **Key Redesign Takeaway:**
  - **Dynamic Situational UX:** Interface navigation hierarchy must dynamically adapt to real-world citizen priorities rather than reflecting static government department structures.

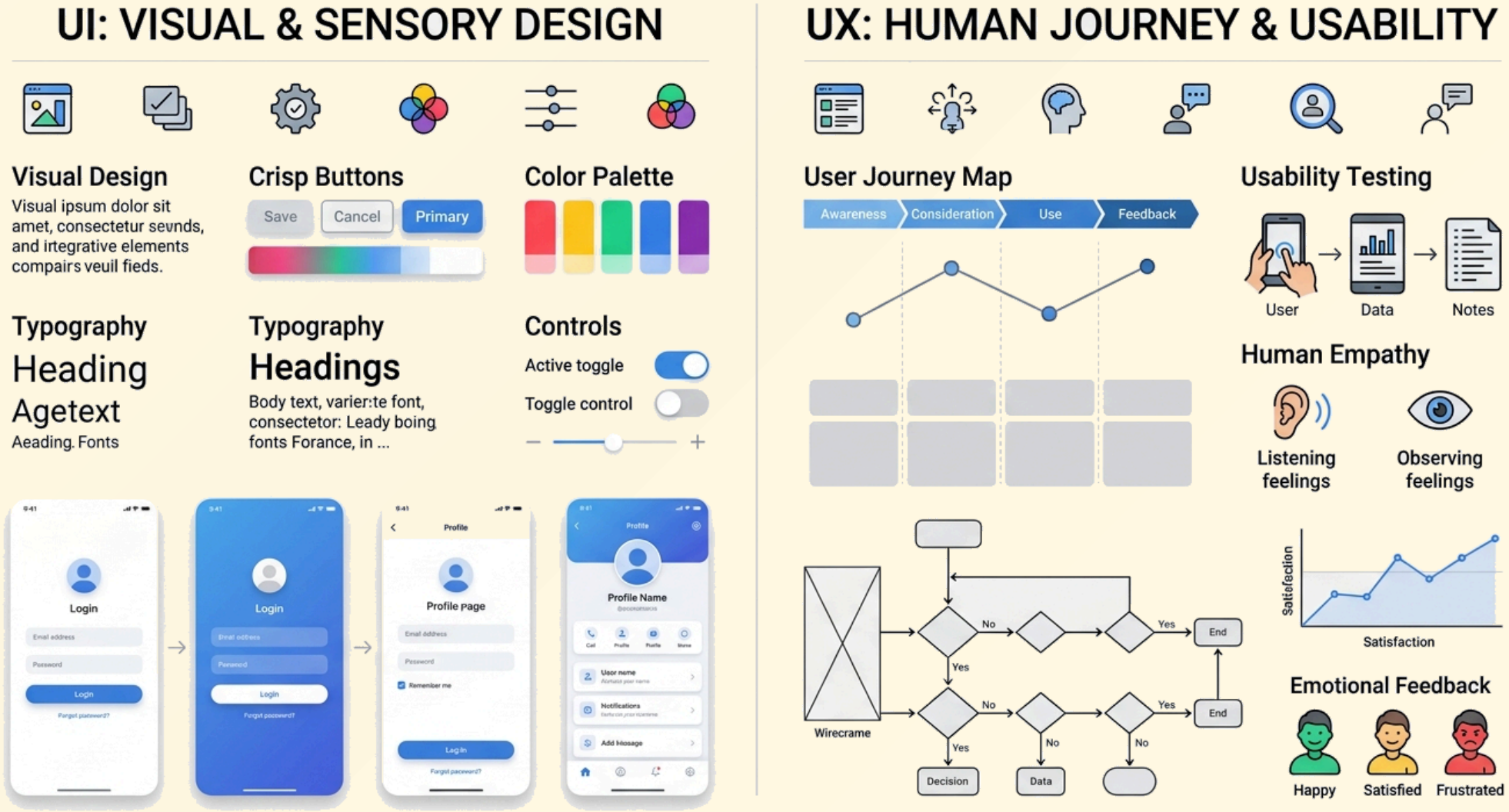
Why is UX So Hard? The Empathy Gap

- **1. "You Are Not the User":**
  - Software engineers possess deep technical mental models (relational schemas, API routes, thread lifecycles).
  - End users possess everyday domain mental models (ordering dinner, booking an appointment, paying a bill).
- **2. The Curse of Knowledge:**
  - Once an engineer builds a complex feature, it seems "obvious" and "intuitive" to them.
  - They forget the cognitive friction experienced by a first-time user encountering the screen cold.
- **3. Defensive vs. Empathetic Thinking:**
  - Engineers often design interfaces to match backend database tables rather than natural human mental steps.

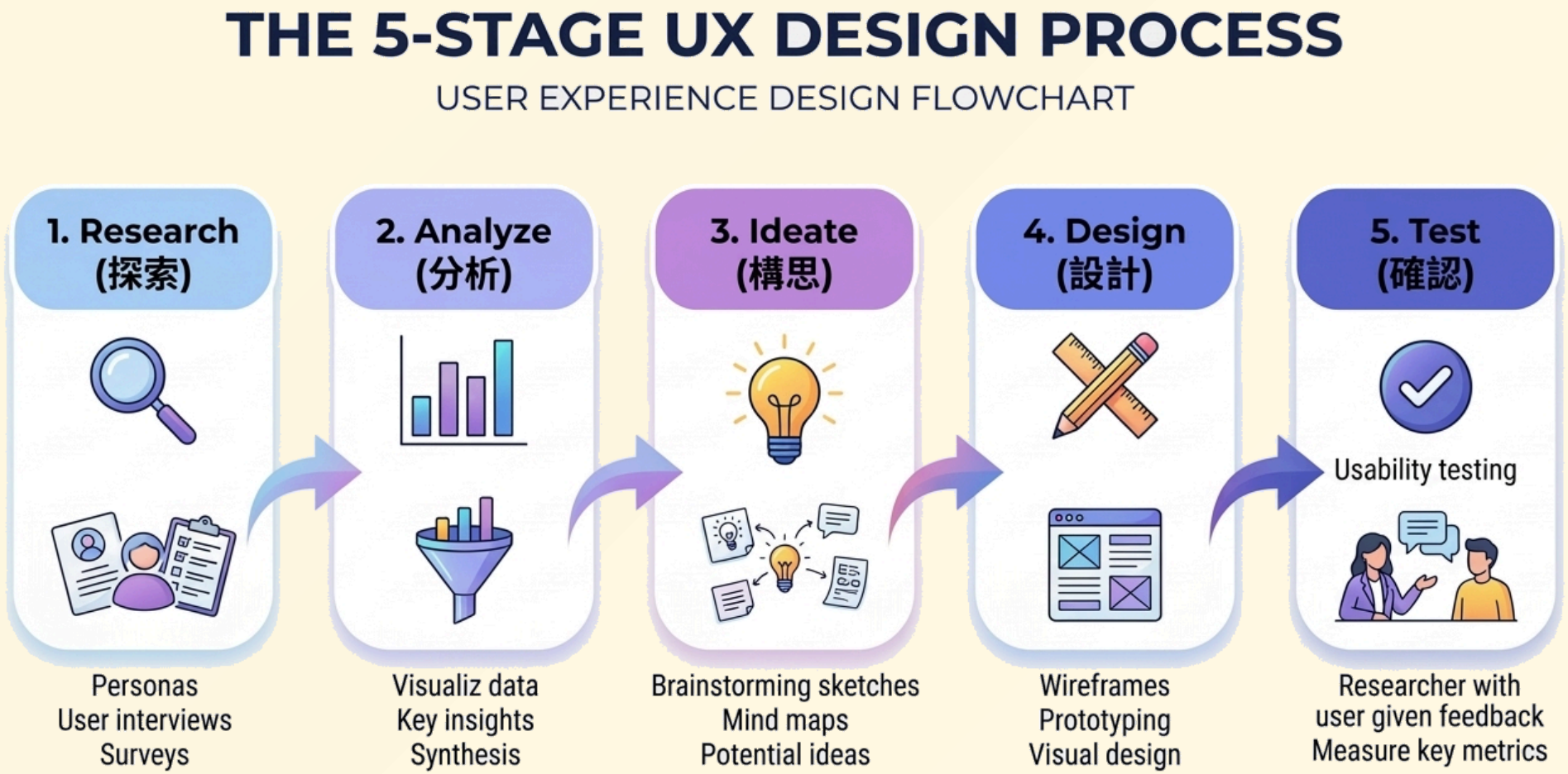
UI (User Interface) vs. UX (User Experience)

- **User Interface (UI):**
  - The tangible, sensory touchpoints through which humans interact with software.
  - *Elements:* Buttons, typography, color palettes, spacing, CSS grids, icons, transitions, animations.
  - *Core Question:* "How does the screen look, render, and react visually?"
- **User Experience (UX):**
  - The overall, end-to-end emotional and psychological journey a human has with a brand, service, or system.
  - *Elements:* Task efficiency, friction, cognitive load, user satisfaction, error recovery, trust.
  - *Core Question:* "How effectively, intuitively, and pleasantly does the user achieve their goal?"
- **The Classical Analogy:**
  - UI is the saddle, stirrup, and reins; UX is the feeling of riding the horse!

Visualizing UI vs. UX: Anatomy of Interaction



The 5-Stage UX Design Thinking Process



The 5 Stages of Design Thinking: Process Breakdown

- **1. Empathize (探索 / 同理):**
  - Conduct qualitative user interviews, contextual field inquiries, and observational studies.
  - Identify unarticulated frustrations and true emotional pain points.
- **2. Define (定義):**
  - Synthesize user research into concrete personas, empathy maps, and precise "How Might We" problem statements.
- **3. Ideate (構思):**
  - Brainstorm diverse potential solutions without premature judgment; sketch rapid low-fidelity paper wireframes.
- **4. Prototype (設計 / 原型):**
  - Build interactive, testable artifacts (clickable Figma mockups, interactive HTML/CSS code components).
- **5. Test (確認 / 測試):**
  - Put prototypes in front of real users, observe stumbling blocks, measure task completion rates, and iterate.

Concept Check Question 1: Taiwan Mask App Usability Audit

During nationwide mask rationing, Taiwan's NHI app originally presented 20 identical square icons in an unprioritized grid. Why did this cause severe operational failure, and how did the redesign resolve it?

- **A.** The interface lacked multi-language localization; adding English and Japanese menus resolved citizen confusion.
- **B.** Equal visual weight buried emergency mask pre-ordering; promoting it to a top Hero Banner aligned interface hierarchy with real-world user priorities.
- **C.** The backend database lacked indexing; adding Redis caching allowed the 20 grid buttons to render twice as fast.
- **D.** Mobile screens should never contain more than 3 buttons; removing all secondary hospital directories solved the problem.

6.2 Nielsen's 10 Usability Heuristics

> *"Jakob's Law: Users spend most of their time on other sites. This means that users prefer your site to work the same way as all the other sites they already know."*
> *— Jakob Nielsen, Nielsen Norman Group*

Overview: Nielsen's 10 Usability Heuristics

> *"Heuristics are broad rules of thumb, not rigid laws—distilled from decades of empirical HCI research."*

- 🚦 **NS01. Visibility of System Status:** Keep users informed through timely feedback.
- 🌍 **NS02. Match System & Real World:** Speak the user's language using natural concepts.
- 🚪 **NS03. User Control & Freedom:** Provide clear emergency exits (undo/redo).
- 📐 **NS04. Consistency & Standards:** Follow platform conventions (Jakob's Law).
- 🛡️ **NS05. Error Prevention:** Eliminate error-prone conditions beforehand.
- 👁️ **NS06. Recognition Rather Than Recall:** Make objects, actions, and options visible.
- ⚡ **NS07. Flexibility & Efficiency:** Provide accelerators for power users.
- 🎨 **NS08. Aesthetic & Minimalist Design:** Maximize signal-to-noise ratio.
- 🩹 **NS09. Help Users Recover from Errors:** Express error messages in plain language.
- 📖 **NS10. Help and Documentation:** Provide searchable, contextual guidance.

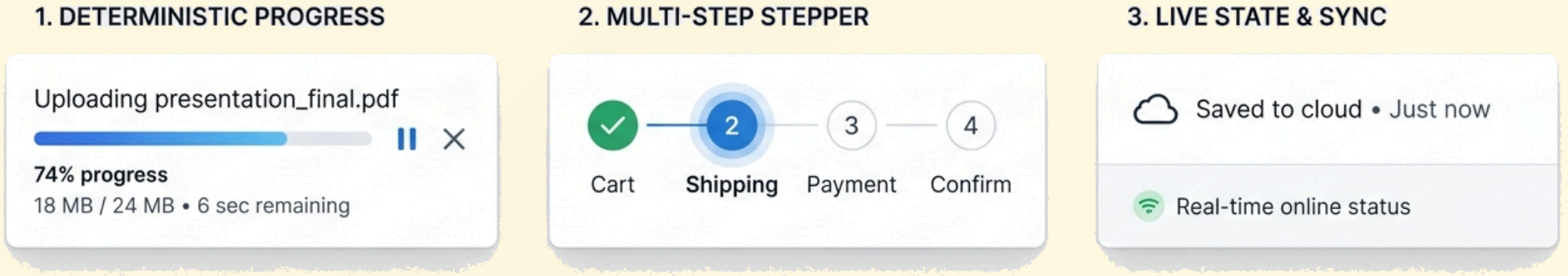
NS01: Visibility of System Status

- **Heuristic Principle:**

> *"The system should always keep users informed about what is going on, through appropriate feedback within a reasonable time."*

- **Core Interaction Patterns:**
  - **Immediate Touch Feedback:** Active button states, ripple effects, and micro-animations validating input registration.
  - **Progress Transparency:** Deterministic progress bars (% complete), multi-step checkout steppers (1. Cart → 2. Shipping → 3. Payment).
  - **Background Task Feedback:** Spinners with descriptive, dynamic status copy ("*Uploading image 3 of 5...*").
  - **Connection State Indicators:** Real-time online/offline banners and live synchronization indicators.

NS01 in Practice: System Status & Real-Time Feedback



NS02: Match Between System and the Real World

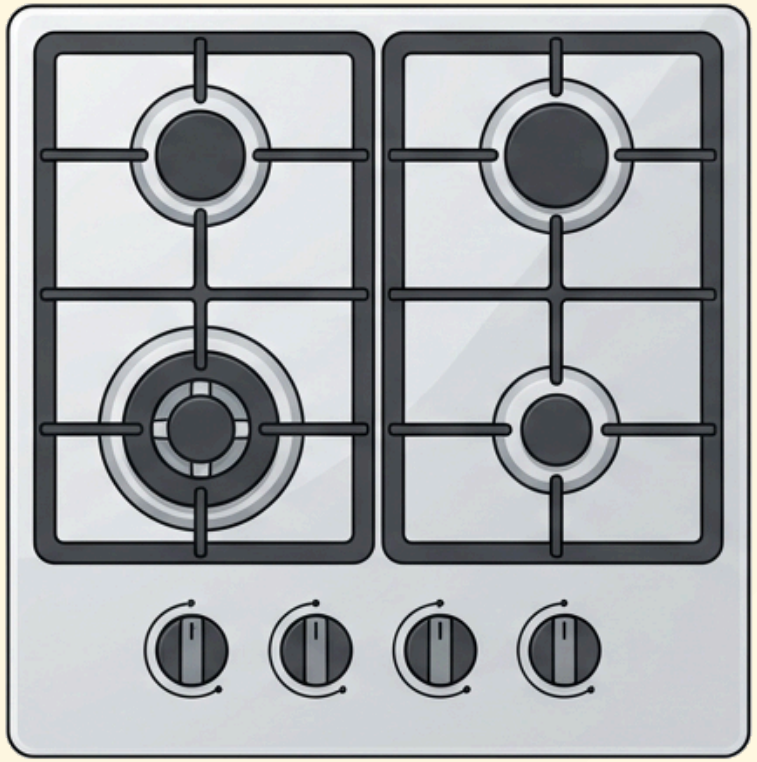
- **Heuristic Principle:**

> *"The system should speak the users' language, with words, phrases, and concepts familiar to the user, rather than system-oriented terms. Follow real-world conventions."*

- **Natural Spatial Mapping (Natural Mapping):**
  - Controls should mirror the physical arrangement of the objects they manipulate.
  - *Classic Counter-Example:* Gas stove burner knobs arranged in a straight line for a 2x2 grid force mental translation and errors.
  - *Good Design:* Arranging 4 knobs in the exact same 2x2 layout as the burners makes operation instantaneous and error-free.
- **Familiar Conceptual Metaphors:**
  - Trash can for deletions, desktop folder icons for directory trees, and shopping carts for e-commerce purchases.

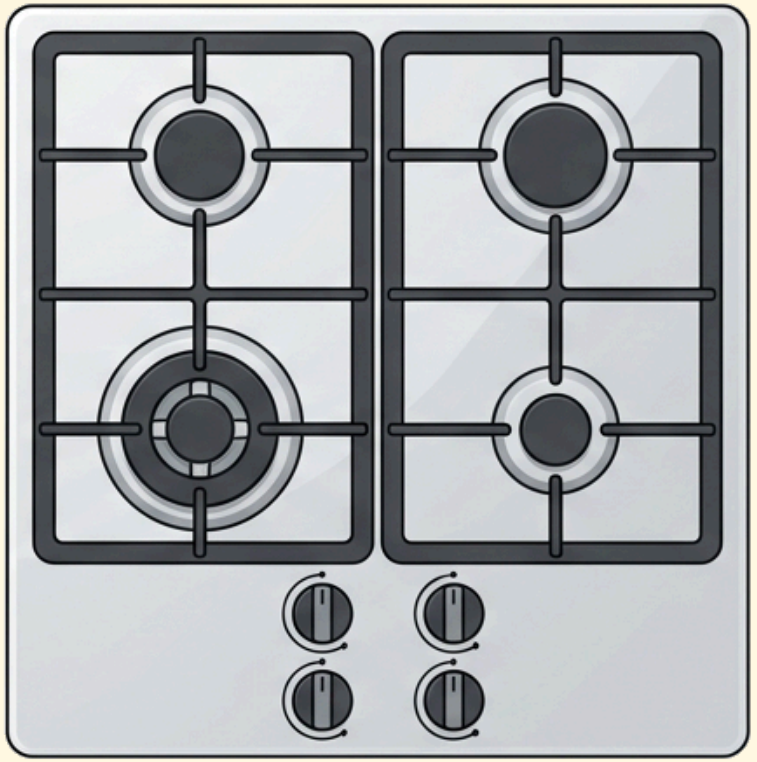
NS02 in Practice: Natural Spatial Mapping of Controls

Unnatural Mapping (Arbitrary)



High Cognitive Friction — Which knob controls which burner? Requires labels and mental translation.

Natural Mapping (Spatial Correspondence)



Intuitive Operation — Subconscious, error-free physical mapping.

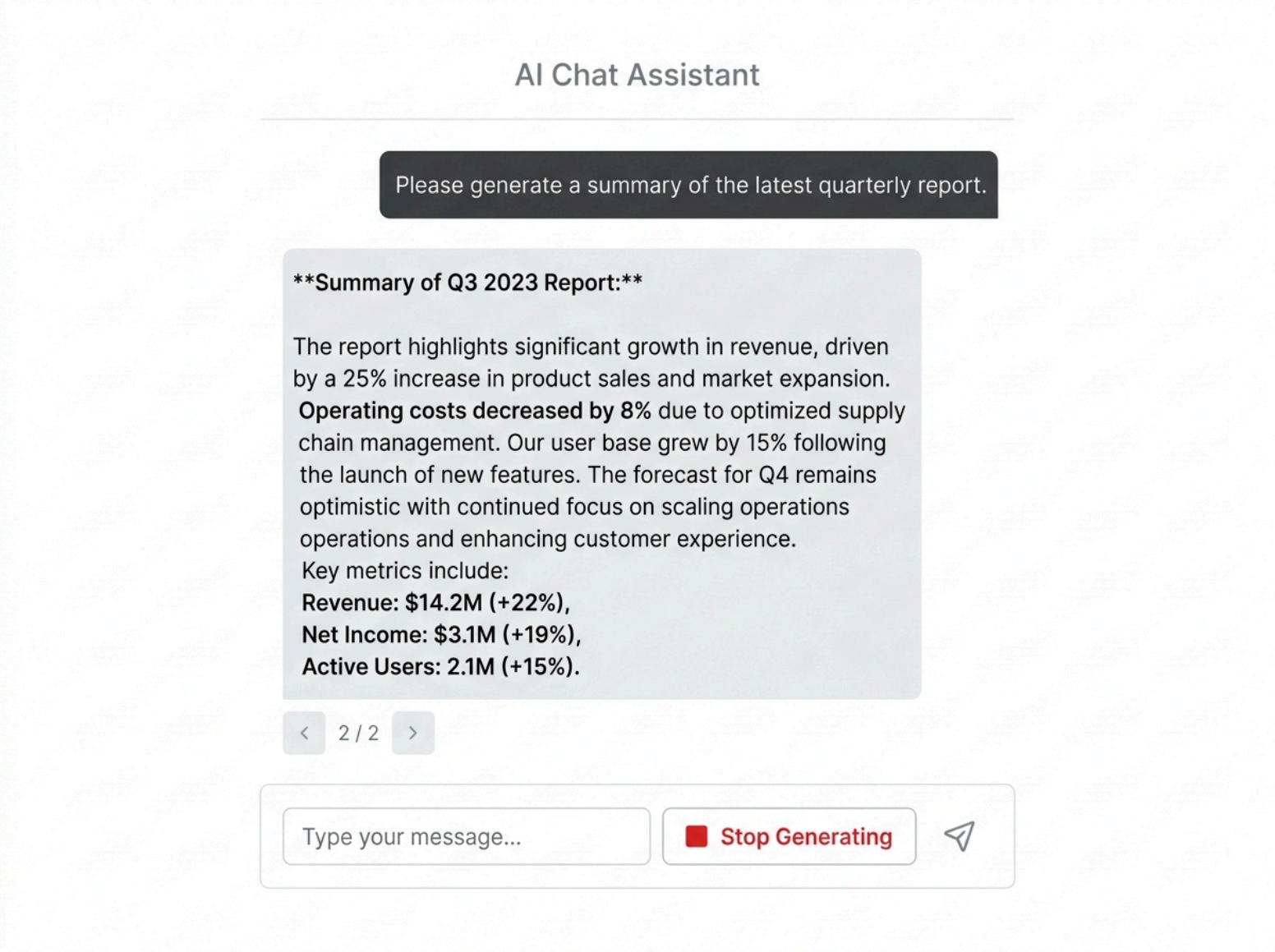
NS03: User Control & Freedom (Emergency Exits)

- **Heuristic Principle:**

> *"Users often choose system functions by mistake and will need a clearly marked 'emergency exit' to leave the unwanted state without having to go through an extended dialogue."*
>
> ”

- **Essential Implementation Patterns:**
  - **Universal Undo / Redo:** Support `Ctrl+Z` / `Cmd+Z` across all text editors, canvas tools, and destructive state changes.
  - **Cancelation Actions:** Provide clear, high-contrast "Cancel" buttons on modals, file uploads, and bulk operations.
  - **Non-Destructive Navigation:** Allow users to back out of a checkout or registration wizard without wiping previously entered form data.
  - **Soft Deletes:** Move deleted items to a "Trash Bin" with a 30-day restoration window before permanent deletion.

NS03 in Practice: User Control, Freedom & Emergency Exits



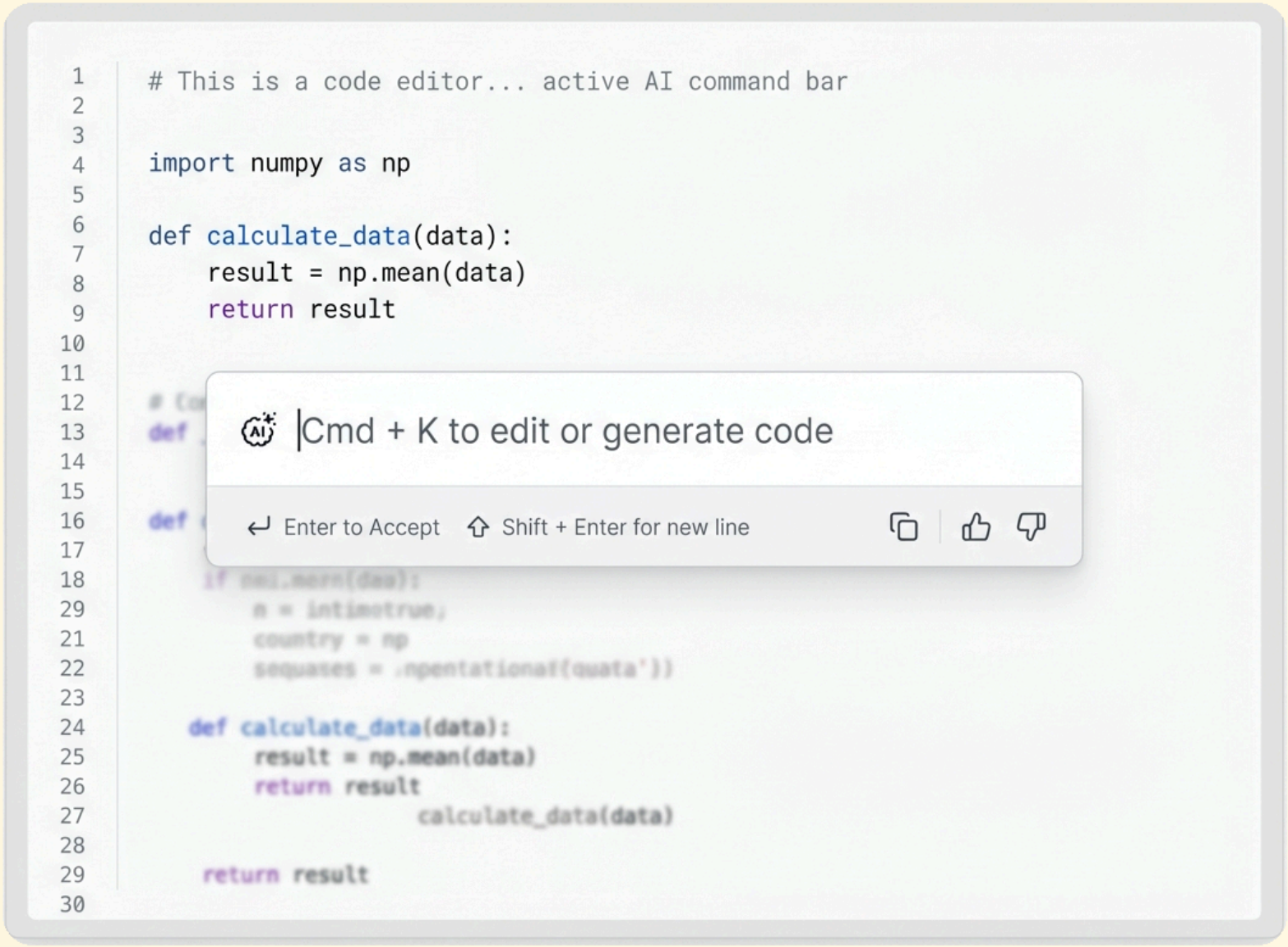
NS04: Consistency & Standards (Jakob's Law)

- **Heuristic Principle:**

> *"Users should not have to wonder whether different words, situations, or actions mean the same thing. Follow platform and industry conventions."*

- **Jakob's Law of Internet User Experience:**
  - Users spend 99% of their digital lives on **other websites and apps**.
  - They expect your software to follow established conventions (e.g., logo at top-left returns home, shopping cart at top-right, magnifying glass indicates search).
- **Two Dimensions of Consistency:**
  - **Internal Consistency:** Uniform fonts, button styles, padding, and terminology across your own app (Design Systems).
  - **External Consistency:** Adhering to platform standards (Apple HIG for iOS, Material Design for Android).

NS04 in Practice: Standardized Conventions & Metaphors



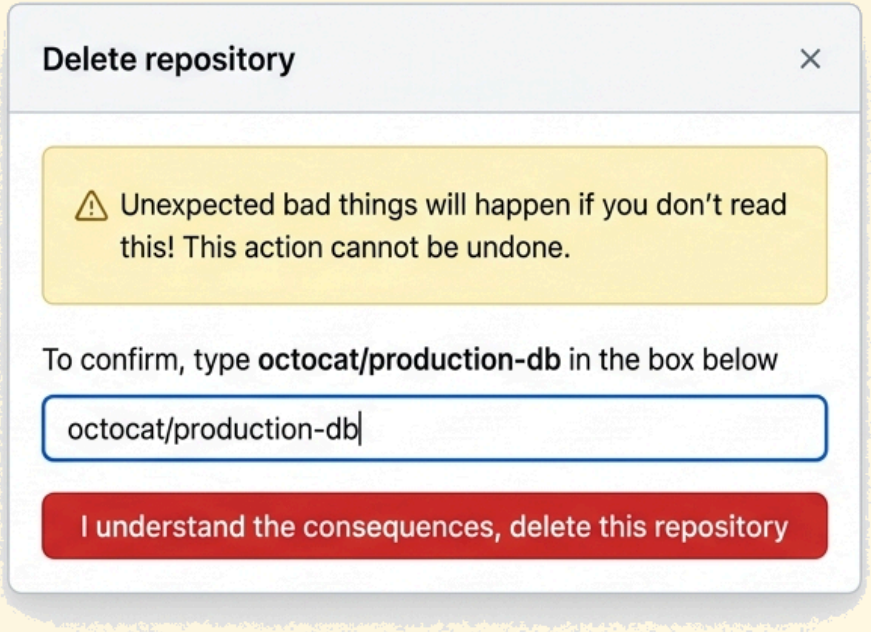
NS05: Error Prevention

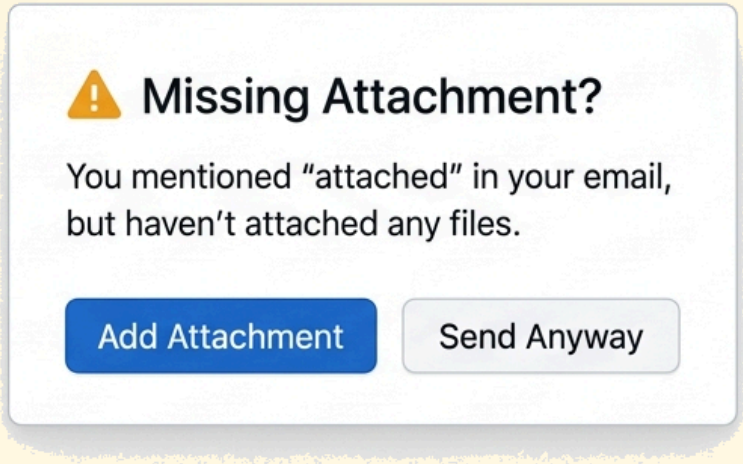
- **Heuristic Principle:**

> *"Even better than good error messages is a careful design which prevents a problem from occurring in the first place."*

- **Defensive Design Patterns:**
  - **Eliminate Error-Prone States:** Grey out or disable the "Submit" button until all mandatory fields pass real-time validation.
  - **Input Constraints:** Use visual Date Pickers instead of freeform text boxes to prevent date formatting syntax errors (`MM/DD/YYYY` vs. `DD/MM/YYYY`).
  - **Confirmation for Irreversible Actions:** Display high-friction confirmation dialogs before permanent actions (e.g., typing the repository name to delete a GitHub repo).
  - **Intelligent Defaults:** Pre-fill sensible default parameters to eliminate blank-form intimidation.

NS05 in Practice: Confirmation Dialogs & Destructive Safeguards





NS06: Recognition Rather Than Recall

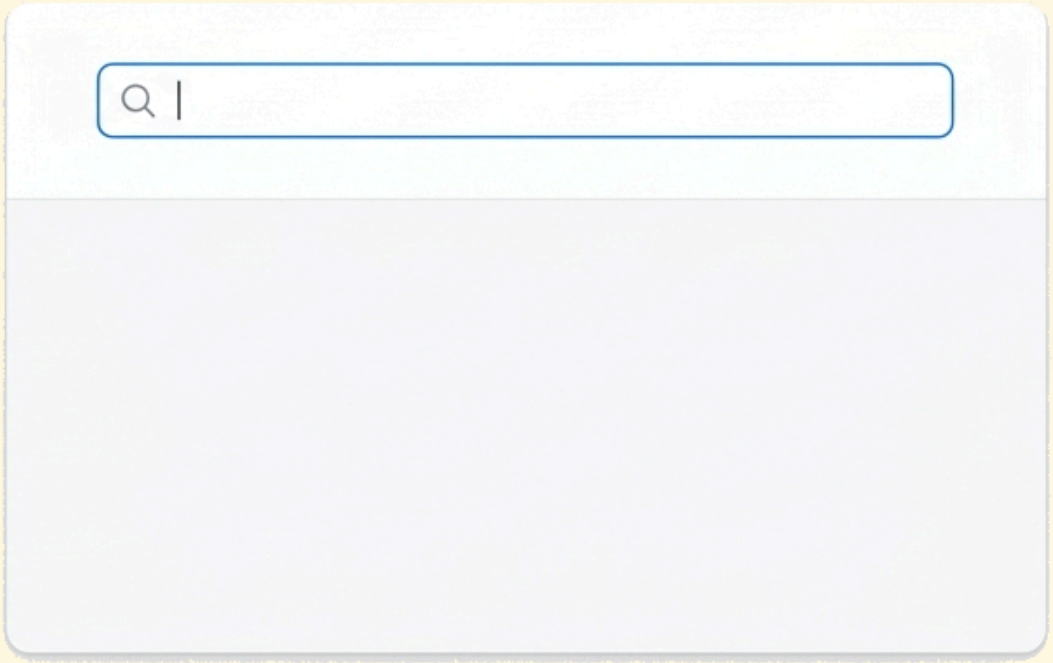
- **Heuristic Principle:**

> *"Minimize the user's memory load by making objects, actions, and options visible. The user should not have to remember information from one part of the dialogue to another."*

- **Cognitive Psychology Foundation:**
  - **Recognition (Easy):** Identifying something you have seen before when presented with visual cues (e.g., multiple-choice quiz).
  - **Recall (Hard):** Retrieving information entirely from memory without assistance (e.g., blank-slate command line terminal).
- **UI Implementations:**
  - Autocomplete search dropdowns displaying recent search keywords.
  - Visual thumbnails of recently edited documents instead of raw filenames.
  - Contextual action toolbars appearing when text is selected.

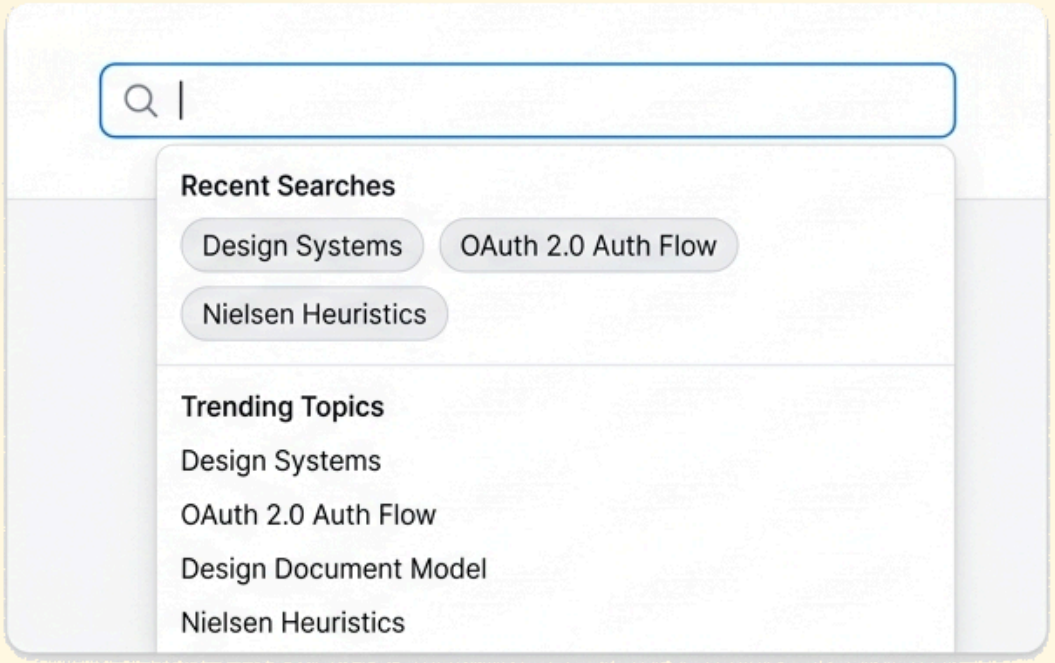
NS06 in Practice: Recognition Over Recall & Search History

Pure Recall (High Cognitive Load)



Forced Recall — Requires mental effort to retrieve past searches from memory.

Effortless Recognition (Low Cognitive Load)



Recognition Over Recall — Past queries and suggestions are immediately visible for 1-tap selection.

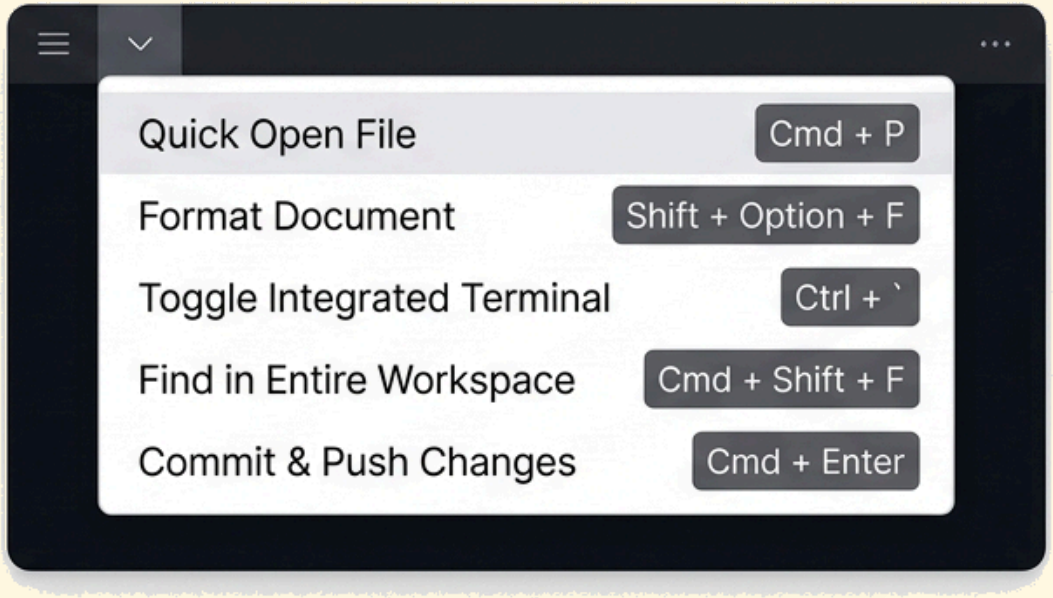
NS07: Flexibility & Efficiency of Use (Accelerators)

- **Heuristic Principle:**

> *"Accelerators—unseen by the novice user—may often speed up the interaction for the expert user such that the system can cater to both inexperienced and experienced users."*
>
> ”

- **Dual-Track Interaction Design:**
  - **Novice Path:** Guided visual menus, onboarding tours, wizard steppers, and descriptive labels.
  - **Expert Accelerators:** Keyboard shortcuts (`Cmd+K` command palette), multi-touch gestures, reusable templates, and batch actions.
- **Progressive Mastery:**
  - Display keyboard shortcut hints directly inside dropdown menus, enabling novices to gradually transition into power users.

NS07 in Practice: Keyboard Shortcuts & Power-User Accelerators



Keyboard Shortcuts & Menu Accelerators

Intelligent Code Snippet Expansion

NS08: Aesthetic & Minimalist Design

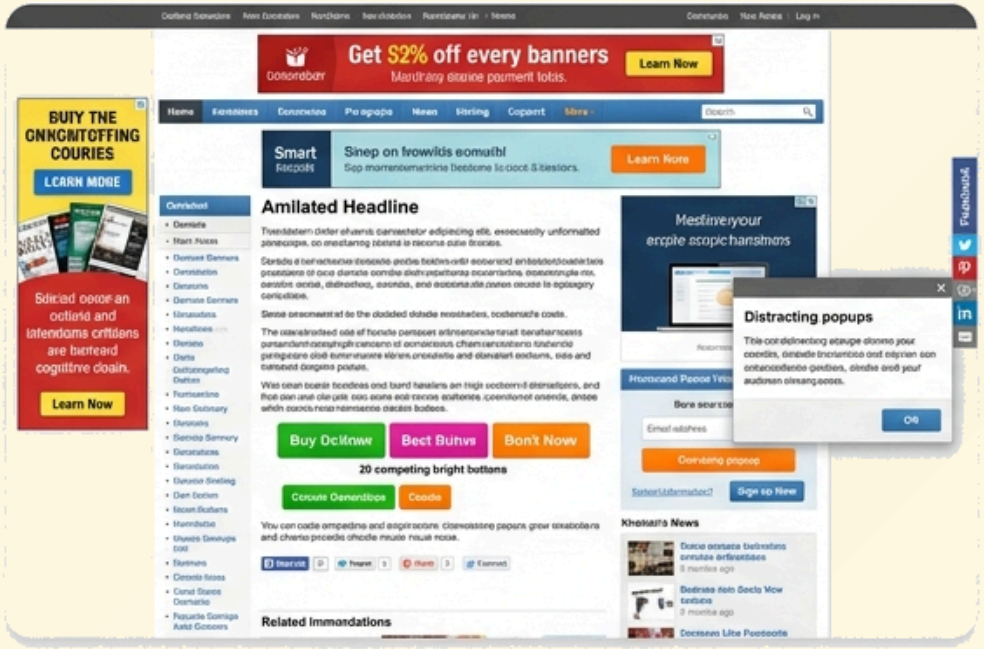
- **Heuristic Principle:**

> *"Dialogues should not contain information which is irrelevant or rarely needed. Every extra unit of information in a dialogue competes with the relevant units of information and diminishes their relative visibility."*

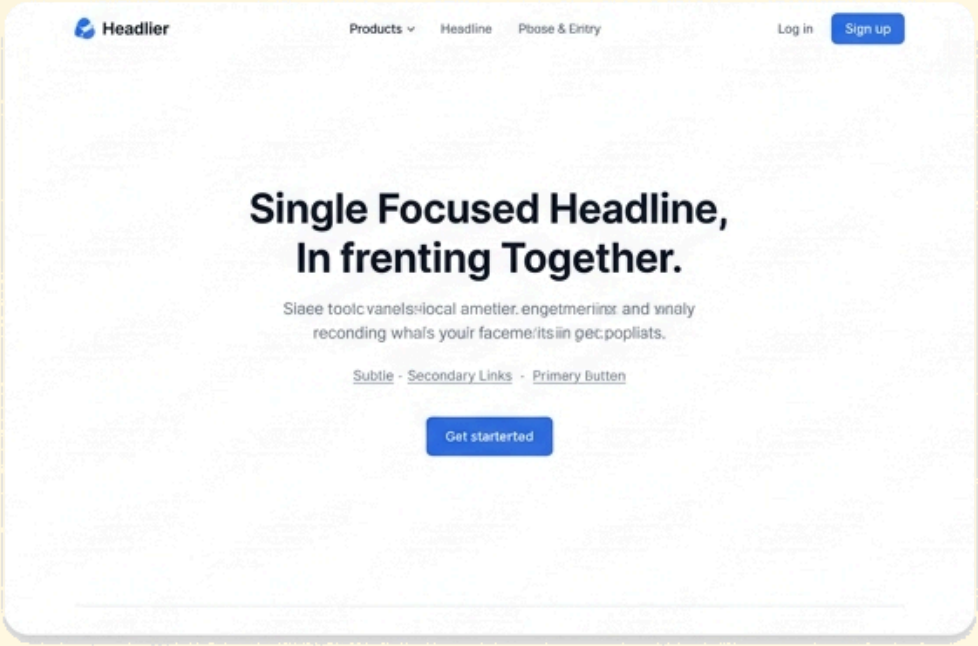
- **The Signal-to-Noise Ratio (SNR):**
  - **Signal:** Information directly relevant to the user's immediate decision.
  - **Noise:** Visual clutter, redundant borders, unnecessary badges, decorative chrome.
- **Technique: Progressive Disclosure (漸進式揭露):**
  - Show only the primary, essential controls upfront.
  - Tuck advanced settings behind "Advanced Options" accordions or secondary tabs.

NS08 in Practice: Minimalist Clarity vs Information Overload

Low Signal-to-Noise Ratio (Information Overload)



High Signal-to-Noise Ratio (Minimalist Clarity)



Visual Noise — Critical actions are buried; high cognitive strain.

Aesthetic Minimalism — Clear visual hierarchy spotlights the primary user goal.

NS09: Help Users Recognize, Diagnose, & Recover from Errors

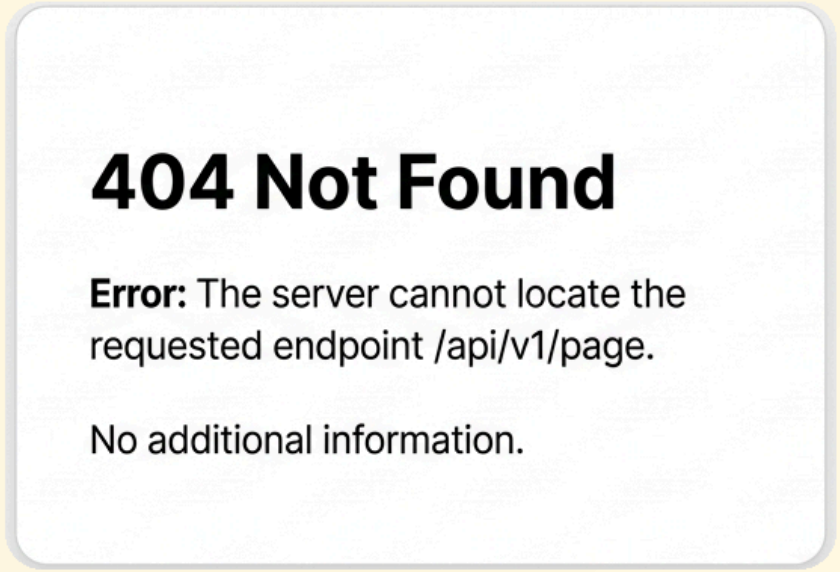
- **Heuristic Principle:**

> *"Error messages should be expressed in plain language (no codes), precisely indicate the problem, and constructively suggest a solution."*

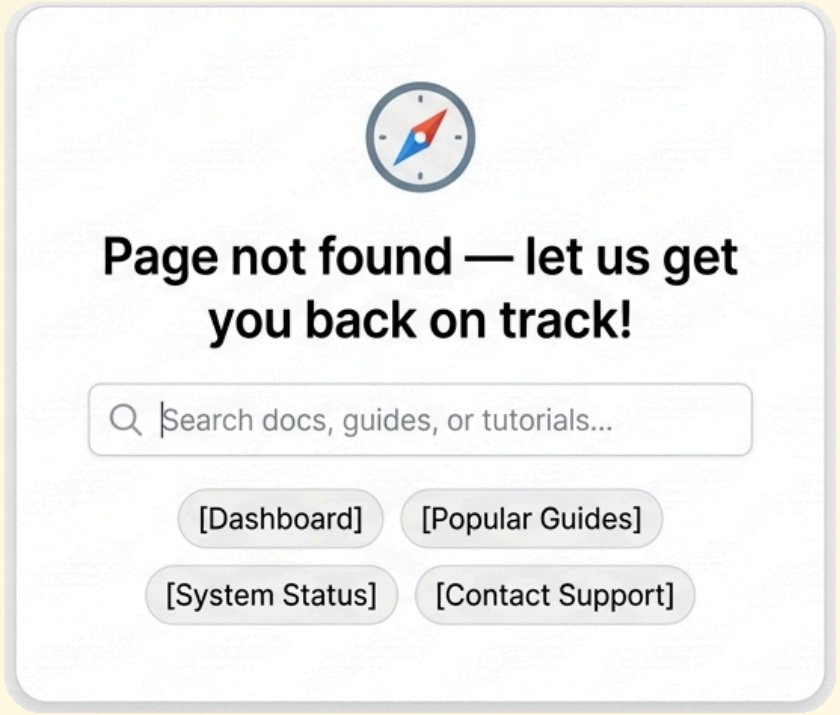
- **The Anatomy of a Terrible Error Message:**
  - ❌ "*Error 0x80004005: Unexpected runtime exception in thread main.*" (Terrifies the user; completely unhelpful).
- **The Anatomy of a Great Error Message:**
  - ✅ **Plain Language:** "*We couldn't charge your credit card.*"
  - ✅ **Precise Cause:** "*The expiration date entered (05/23) is in the past.*"
  - ✅ **Constructive Recovery:** "*Please enter an updated expiration date or choose a different payment method [Update Card].*"

NS09 in Practice: Constructive Error Recovery & Helpful 404s

Cryptic Error (Dead End)



Constructive Recovery (Helpful 404)



NS10: Help and Documentation

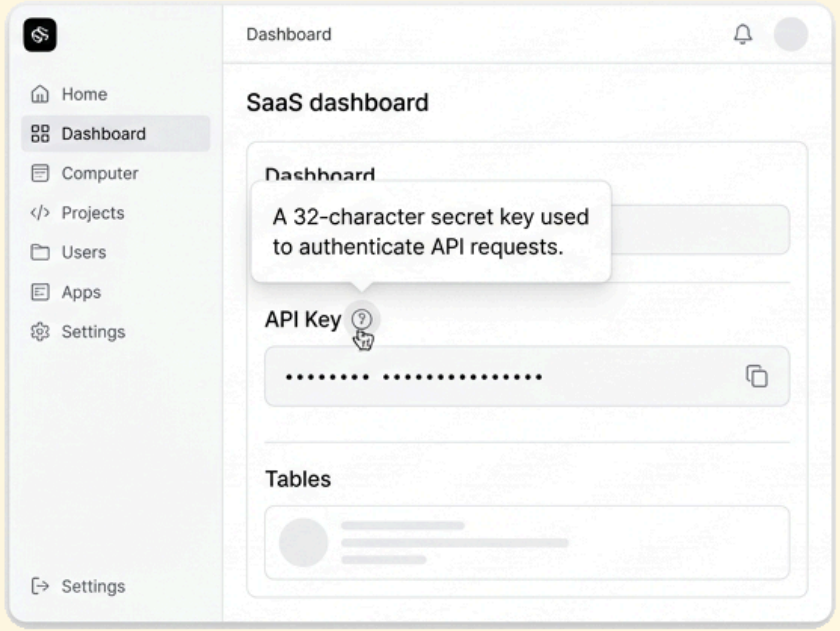
- **Heuristic Principle:**

> *"Even though it is better if the system can be used without documentation, it may be necessary to provide help and documentation. Any such information should be easy to search, focused on the user's task, list concrete steps to be carried out, and not be too large."*

- **Modern Documentation Paradigms:**
  - **Contextual Tooltips:** Inline `?` icons explaining complex form requirements on hover.
  - **Interactive Onboarding Tours:** Lightweight, multi-step interactive walkthroughs highlighting key UI areas for first-time signups.
  - **Searchable Task-Oriented Help Centers:** Documentation organized by customer goals rather than internal system architecture.

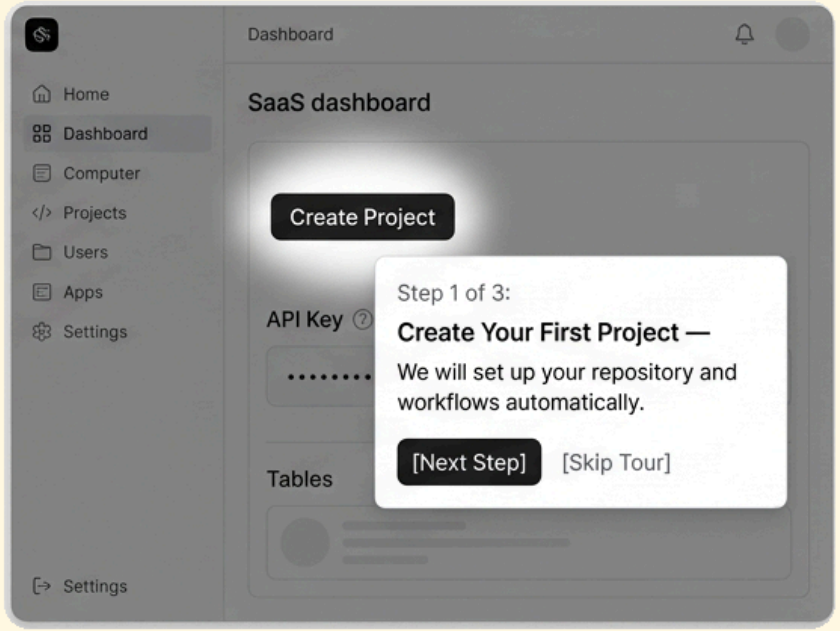
NS10 in Practice: Contextual Help & Interactive Onboarding

Passive
Contextual Help



Unobtrusive — Accessible on demand without cluttering the screen.

Active
Interactive Onboarding



Concept Check Question 2: Usability Heuristics

When designing an e-commerce checkout flow, a team replaces raw text date inputs with an interactive calendar picker and automatically disables the "Place Order" button until all required shipping addresses pass validation. Which Nielsen Heuristic is primarily demonstrated?

- **A.** NS01: Visibility of System Status.
- **B.** NS05: Error Prevention.
- **C.** NS07: Flexibility and Efficiency of Use.
- **D.** NS10: Help and Documentation.

6.3 AI for UX (AI4UX): Amplifying Design Workflows

> *"AI will not replace UX designers, but UX designers who use AI will replace those who do not."* „

What is AI for UX (AI4UX)?

- **The AI-Amplified Designer:**
  - Generative AI acts as a **cognitive partner and speed multiplier** across every stage of the design thinking process.
- **Core Application Areas:**
  - **Research & Empathy:** Generating diverse synthetic user personas, simulating edge-case user interviews.
  - **Ideation & Copywriting:** Rapidly generating microcopy variations, localization strings, and accessibility alt text.
  - **Automated Heuristic Audits:** Feeding UI screenshots to multimodal LLMs to identify Nielsen heuristic violations.
  - **Generative Prototyping:** Text-to-wireframe and text-to-code pipelines (v0, Figma AI, Galileo).

AI for UX: The Augmented Human-AI Design Workflow

The RTCF Prompt Engineering Framework for Designers

To get precise, professional UX output from AI models, avoid vague prompts (❌ "*Help me design a login page*"). Use the structured **RTCF Framework**:

- **R — Role (角色):**
  - Define the AI's persona and expertise (e.g., "*Act as a Lead UX Researcher specializing in accessibility and WCAG 2.1 AA standards.*").
- **T — Task (任務):**
  - Define the concrete objective (e.g., "*Conduct a heuristic evaluation of this mobile checkout wireframe.*").
- **C — Constraints (約束):**
  - Enforce domain rules (e.g., "*Evaluate strictly against Nielsen's Heuristics NS01, NS03, and NS05; target a 375px mobile viewport.*").
- **F — Format (格式):**
  - Specify the output structure (e.g., "*Provide results in a 4-column Markdown table: [Heuristic ID, Observed Issue, Severity (0-4), Concrete Fix Recommendation].*").

Multi-Agent Prototyping & Generative UI

- **Autonomous Multi-Agent Architecture:**
  - Modern IDEs and agent systems (e.g., Antigravity, Devin, Claude Code) orchestrate specialized agents working in tandem.
  - **UX Research Agent:** Analyzes customer tickets and synthesizes user pain points.
  - **Design System Agent:** Generates CSS tokens, typography scales, and accessible color palettes.
  - **Frontend Code Agent:** Translates specifications into responsive HTML/CSS/React components.
- **The Human-in-the-Loop Safeguard:**
  - AI generates rapid prototypes; human engineers evaluate ergonomics, brand voice, and real user empathy.

Multi-Agent Architecture: Collaborative Prototyping Pipelines

Google Antigravity Agent

Modern AI Pair-Programming & Autonomous Engineering Engine

Read Every Turn • Permanent Full Injection

📋 AGENTS.md (Project Directives & Rules)

- **Hierarchical Scope Traversal:** Automatically walks up directories from local folder to repository root and global configs.
- **Enforces Engineering Standards:** Standardizes naming conventions, tech stacks, and domain terminology across the team.
- **Core UX & Architecture Guardrails:** Hard rules for real-time status feedback, destructive action safeguards, and zero broken links.
- **Safeguards Design Consistency:** Enforces unified design tokens, typography, and interactive component behaviors (NS04).

Task Match Triggered • Progressive Disclosure

🧰 .agents/ (Modular Skills & Tools)

- **Lightweight YAML Headers:** Only skill descriptions are indexed initially, saving valuable context window tokens.
- **Dynamic Intent Activation:** Full SOPs, reference manuals, and execution steps are loaded on demand upon task match.
- **Encapsulated Executable Scripts:** Bundles local automation scripts (`scripts/`) for autonomous loop execution.
- **Deep Ecosystem Integration:** Native support for extensible Model Context Protocol (MCP) servers and lifecycle workflow hooks.

6.4 UX for AI (UX4AI): Designing Human-Centered AI

> *"The hardest part of building AI products is not training the model—it is designing the interface that makes probabilistic intelligence understandable, steerable, and trustworthy."*

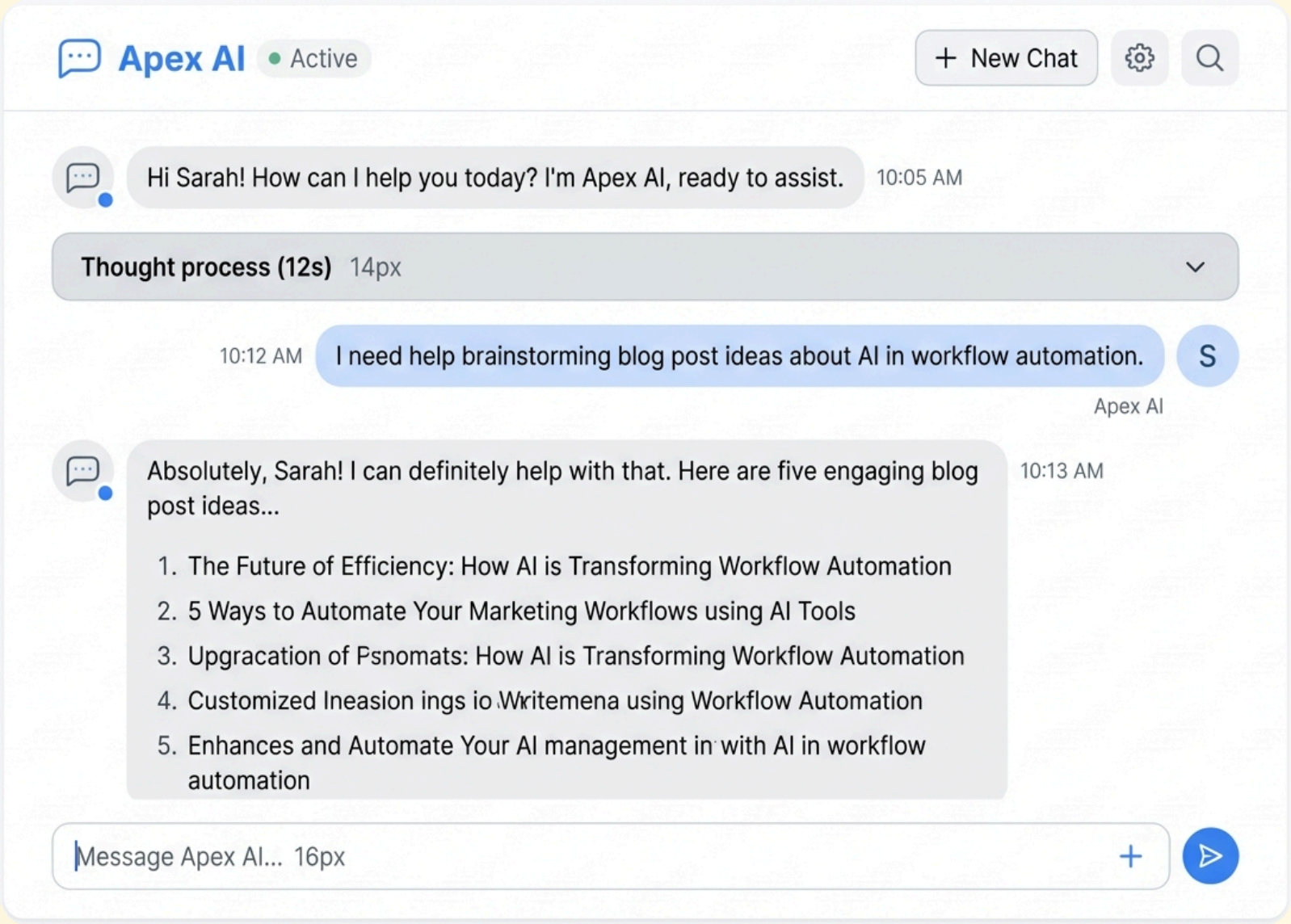
Unique UX Challenges in AI-Native Products

- **1. Non-Determinism & Unpredictability:**
  - Traditional software: `2 + 2` always returns `4`.
  - AI systems: Submitting the identical prompt twice may yield completely different phrasing, tone, or formatting.
- **2. The "Hallucination" Dilemma:**
  - Generative models can generate completely fabricated facts, non-existent API methods, or fake citations with total syntactic confidence.
- **3. Latency & the "Thinking" Black Box:**
  - Complex reasoning models (e.g., o1, Gemini Thinking) require 5 to 30 seconds of inference time before returning a single token.
- **4. The Calibration of Trust:**
  - **Over-trust:** Users blindly accept false AI code without reviewing it, leading to production outages.
  - **Under-trust:** Users abandon the AI after a single minor mistake, losing all productivity benefits.

AI + NS01: Visibility of AI State & Thinking Traces

- **The Problem of AI Latency & Black Boxes:**
  - Long inference times (10-30s) leave users staring at blank screens, wondering if the server crashed.
- **Interaction Solutions:**
  - **Streaming Tokens:** Stream text responses token-by-token immediately, providing instant perceived performance.
  - **Collapsible Reasoning Traces:** Display interactive "Thinking Process" accordions showing internal reasoning steps and tool execution.
  - **RAG Grounding Citations:** Provide clickable source pills linking directly to retrieved reference documents.

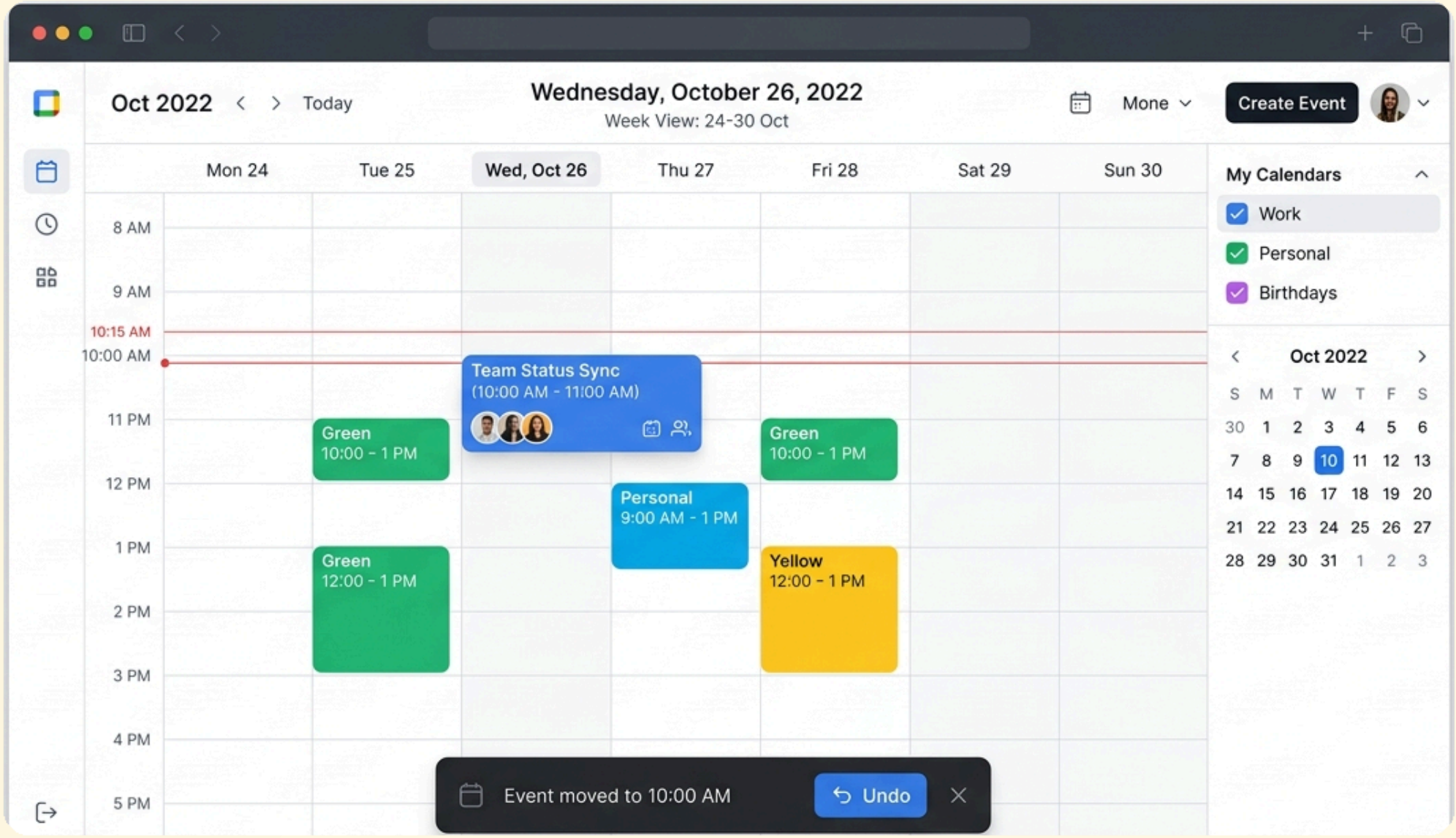
AI + NS01 in Practice: Streaming Tokens & Collapsible Reasoning



AI + NS02 & NS03: Grounding, Control & Human Agency

- **AI + NS02 (Match with Real World):**
  - **Transparent Bot Identity:** Clearly disclose that the system is an AI assistant; avoid deceptive pseudo-human personas.
  - **Conversational Grounding:** Establish shared context using familiar conversational turns and natural language.
- **AI + NS03 (User Control & Freedom):**
  - **Stop Generation Button:** Provide an instant button to abort runaway or misdirected responses.
  - **Prompt Branching:** Allow users to edit previous prompts and fork alternative conversation threads.
  - **Steerability Sliders:** Provide controls for temperature, brevity, and response tone.

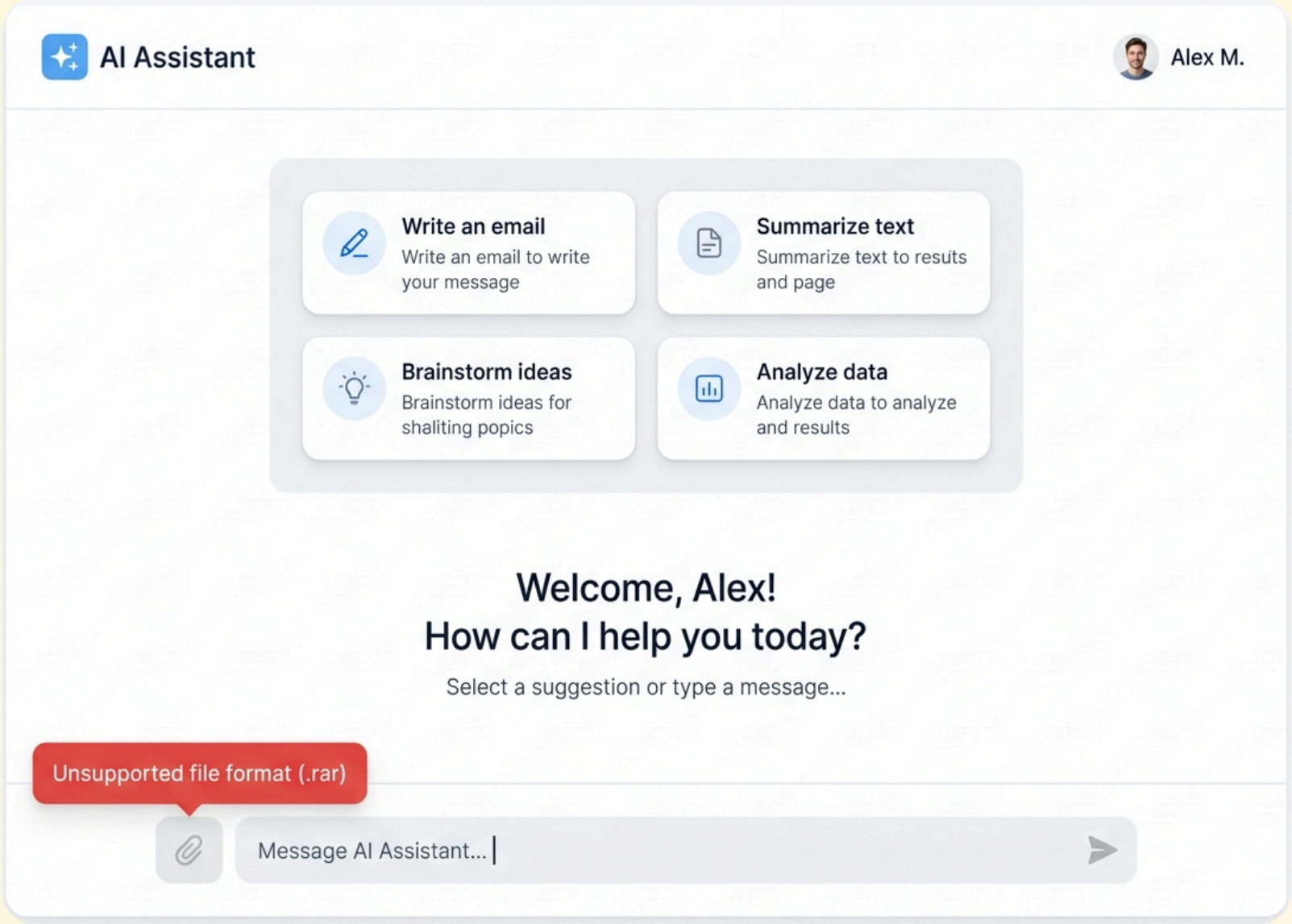
AI + NS03 in Practice: User Steerability & Cancelation Controls



AI + NS05: Error Prevention & Agentic Guardrails

- **The Extreme Stakes of Agentic AI:**
  - Autonomous agents can execute shell commands, edit production code, and interact with external APIs.
  - An unchecked hallucination can delete files, drop database tables, or leak API keys.
- **Human-in-the-Loop (HITL) Guardrails:**
  - **Pre-Execution Confirmation:** Require explicit developer approval before running terminal commands or modifying files.
  - **Diff Previews:** Display color-coded git-style diffs showing exactly what lines the AI proposes to modify.
  - **Sandboxed Execution:** Run unverified AI code in isolated containers before applying changes to host environments.

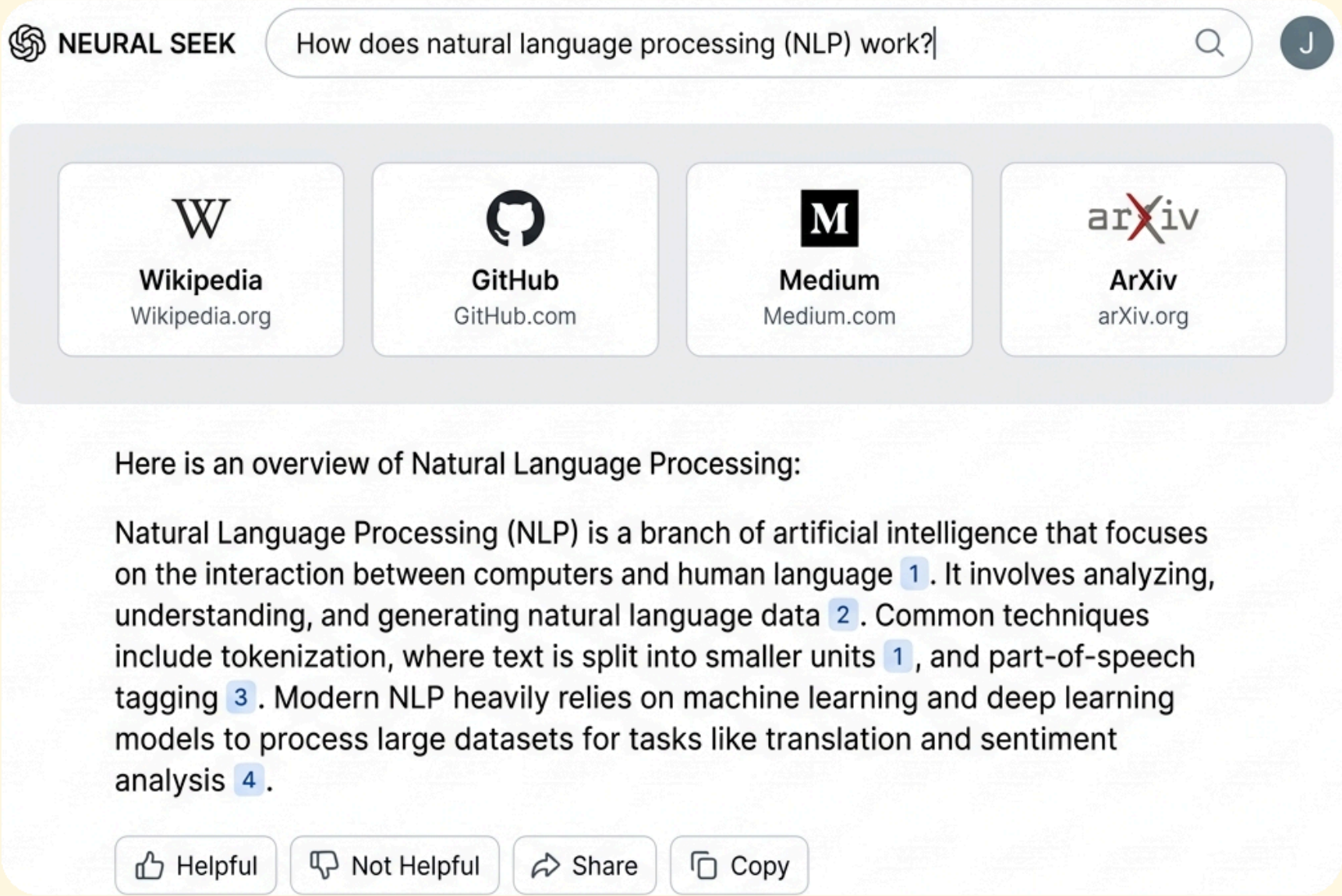
AI + NS05 in Practice: Human-in-the-Loop Confirmation & Diff Auditing



AI + NS07 & NS10: Accelerators & Explainability (XAI)

- **AI + NS07 (Flexibility & Accelerators):**
  - **Slash Commands:** Quick keyboard triggers (`/explain`, `/refactor`, `/fix-tests`) allowing power users to bypass conversational typing.
  - **Multi-Modal Drag-and-Drop:** Allowing users to drag screenshots, PDF diagrams, and log files directly into the prompt box.
- **AI + NS10 (Explainable AI - XAI):**
  - **"Why this answer?":** Contextual tooltips explaining which source documents or prompt instructions triggered the recommendation.
  - **Model Cards & Confidence Meters:** Explicitly disclosing active model versions and estimated prediction confidence.

AI + NS10 in Practice: Explainable AI & Contextual Citations



Concept Check Question 3: UX for AI Design

When designing an autonomous AI coding assistant that can inspect directories, edit source files, and execute shell commands, which design pattern best fulfills **Error Prevention (NS05)** and **User Agency (NS03)**?

- **A.** Allowing the agent to execute all file deletions and shell scripts silently in the background without user interruption.
- **B.** Requiring explicit human-in-the-loop review and confirmation before executing high-risk file modifications or system terminal commands.
- **C.** Removing the ability to stop or cancel running background tasks once the AI begins executing code.
- **D.** Hiding all command execution logs from developers to avoid visual clutter and information overload.

6.5 Evaluation & Engineering Synthesis

> *"Evaluating usability with 5 users uncovers roughly 85% of all usability problems."*
> *— Jakob Nielsen*

Conducting a Formal Heuristic Evaluation

- **What is a Heuristic Evaluation?**
  - A discount usability inspection method where independent evaluators examine an interface against established heuristics (e.g., Nielsen's 10).
- **Nielsen's 5-Point Severity Rating Scale:**
  - **0 = Not a usability problem:** Does not affect user progress.
  - **1 = Cosmetic problem only:** Minor visual blemish; fix only if extra time is available.
  - **2 = Minor usability problem:** Low priority; causes minor inconvenience but user can recover.
  - **3 = Major usability problem:** High priority; severely delays or confuses users; must be fixed before release.
  - **4 = Usability catastrophe:** Imperative to fix! Completely blocks task completion; software cannot ship!

Heuristic Evaluation vs. Empirical User Testing

Dimension	Heuristic Evaluation (專家啟發式評估)	Empirical User Testing (真實使用者易用性測試)
Evaluators	3 to 5 UX Experts / Senior Engineers	5 Real Representative End-Users
When to Use	Early in design (wireframes, early prototypes)	Later in development (interactive prototypes, staging)
Cost & Speed	Extremely fast (1–2 days); low financial cost	Slower (1–2 weeks); requires recruitment and lab setup
Primary Strength	Quickly finds obvious heuristic violations & edge cases	Reveals authentic human behavioral friction & surprises
Limitation	Experts can miss domain-specific workflow quirks	Users reveal *what* broke, but not always *why* or *how to fix*

Chapter Recap: Fill-in-the-Blank Challenge

Test your mastery of the core concepts in Chapter 6:

- **1.** A door with a pull handle that must be pushed to open is famously known as a ____ Door.
- **2.** ____ Law states that users spend most of their time on other websites, expecting your site to follow common conventions.
- **3.** Nielsen Heuristic ____ states that preventing mistakes beforehand is vastly superior to displaying polite error messages.
- **4.** The prompt engineering framework for designers consisting of Role, Task, Constraint, and Format is abbreviated ____.
- **5.** In AI interfaces, displaying streaming tokens and collapsible reasoning steps exemplifies heuristic ____: Visibility of System Status.
- **6.** On Nielsen's severity rating scale, a usability defect that completely blocks task completion and prevents product release is rated as a ____.

Chapter Recap: Solution Key

Here are the completed principles and technical terms:

- **1.** A door with a pull handle that must be pushed to open is famously known as a **Norman** Door.
- **2. Jakob's** Law states that users spend most of their time on other websites, expecting your site to follow common conventions.
- **3.** Nielsen Heuristic **NS05: Error Prevention** states that preventing mistakes beforehand is vastly superior to displaying polite error messages.
- **4.** The prompt engineering framework for designers consisting of Role, Task, Constraint, and Format is abbreviated **RTCF**.
- **5.** In AI interfaces, displaying streaming tokens and collapsible reasoning steps exemplifies heuristic **NS01**: Visibility of System Status.
- **6.** On Nielsen's severity rating scale, a usability defect that completely blocks task completion and prevents product release is rated as a **4 (Catastrophe)**.

Chapter Summary & Engineering Takeaways

- **1. Human Psychology Precedes Code:**
  - Excellent software engineers design for human cognitive limits (working memory, emotional frustration, mental models).
  - When users stumble, fix the interface affordances rather than blaming user competence.
- **2. Internalize Nielsen's 10 Heuristics:**
  - Make status visible (NS01), match real-world metaphors (NS02), offer undo exits (NS03), adhere to standards (NS04), prevent errors proactively (NS05), and write constructive error messages (NS09).
- **3. Master AI-Augmented Design (AI4UX & UX4AI):**
  - Use structured RTCF prompting to accelerate research, prototyping, and automated heuristic audits.
  - Design AI products for calibrated trust: stream tokens, show reasoning traces, mandate human confirmations for agentic actions, and provide in-place error recovery.